

Business Process Management

Campus Enablement Program: Day 2
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MakeLabs.in



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Agenda for Day 2

- Designing a process
- Configuring Basic Elements and Shapes
 - Boundaries
 - Events
 - Connectors
- DMN Overview
- FEEL: Expression Language
- DMN Advanced Elements

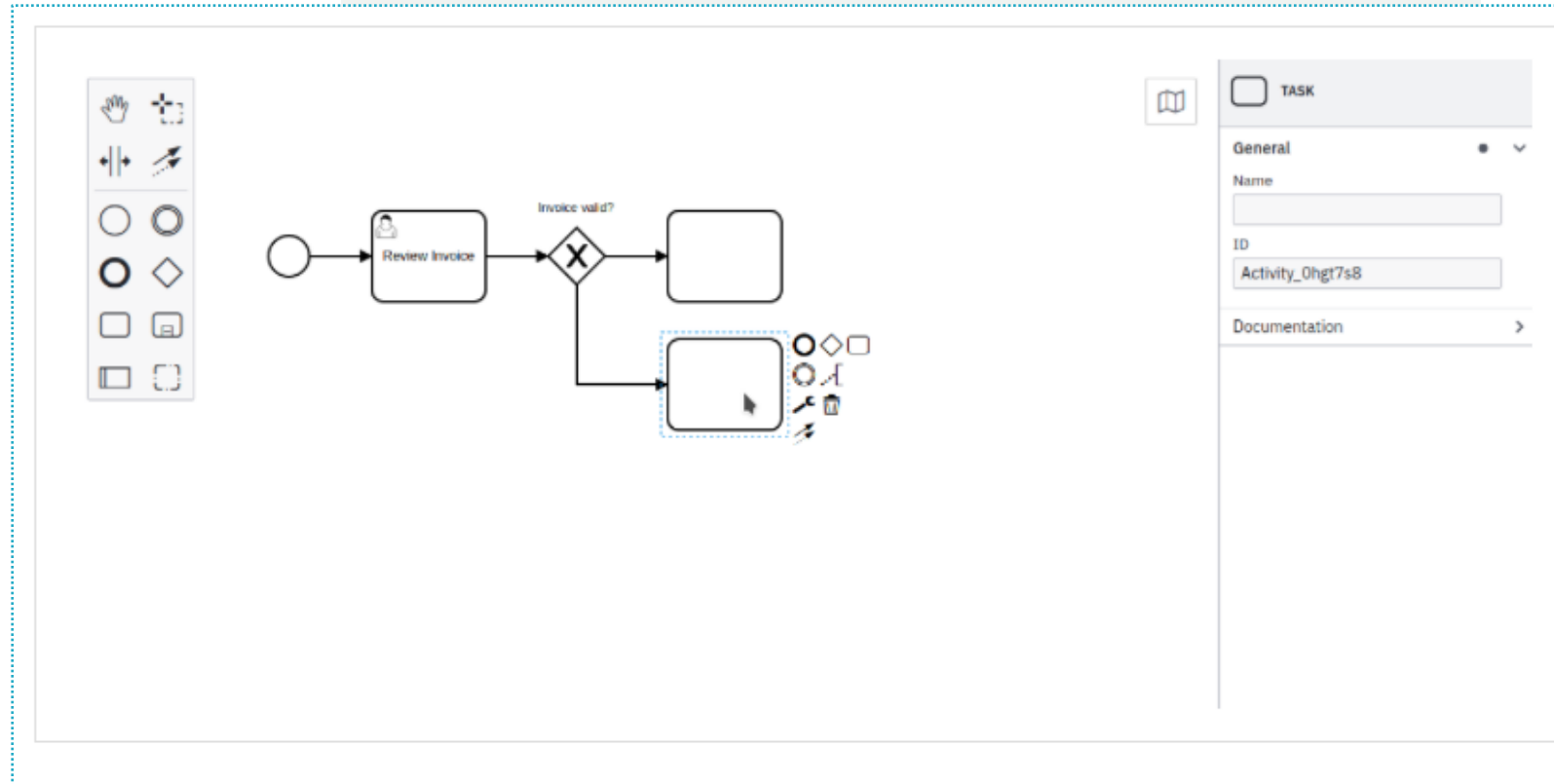


Designing a Business Process

Business Process Model and Notation (**BPMN**) was developed as a graphical notation to represent complex processes.

It is maintained by the non-profit The Object Management Group (**OMG**) and employed by numerous organizations globally.

The visual nature of BPMN enables greater collaboration between different teams, particularly within **Modeler**.

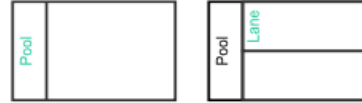


Designing a Business Process

Basic Elements

- Swimlane
- Subprocess
- Tasks
- Gateways
- Markers

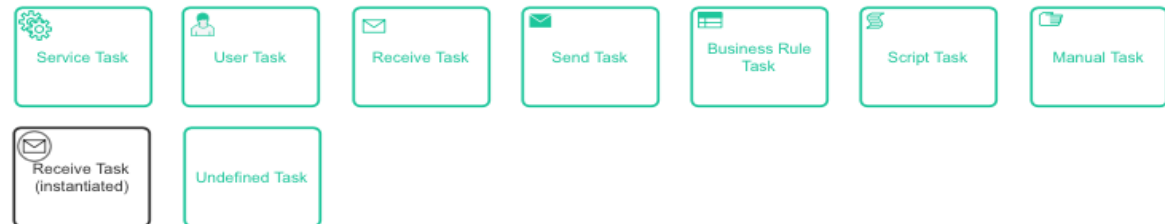
Participants



Subprocesses



Tasks



Gateways



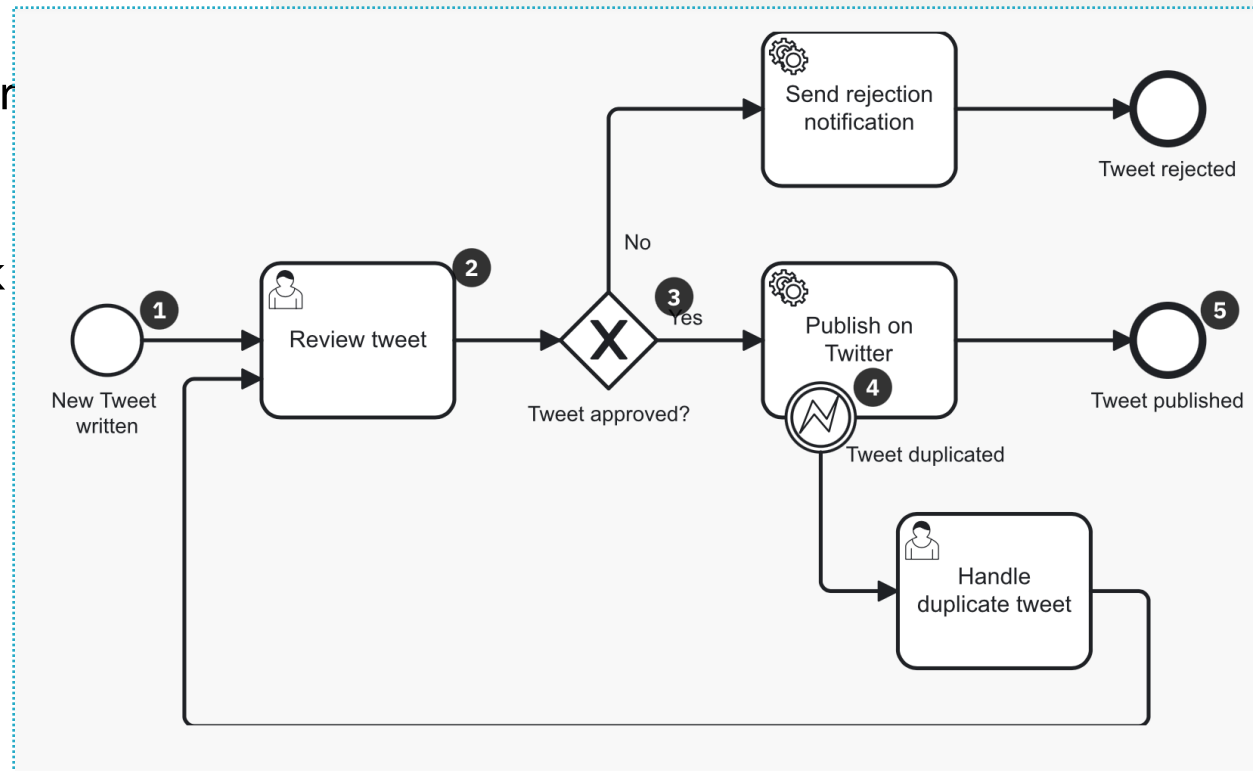
Markers



Designing a Business Process

How to make it readable?

- 1) Start event labels informs the reader of how the process is triggered.
- 2) An activity - labeled as "activity" - informs the reader of the piece of work to be carried out.
- 3) Gateway labels clarifies based on which condition(s) and along which sequence flow the process proceeds.
- 4) Labeled boundary events clearly express in which cases a process execution might follow an exceptional path.
- 5) Labeled end events characterize end results of the process from a business perspective.



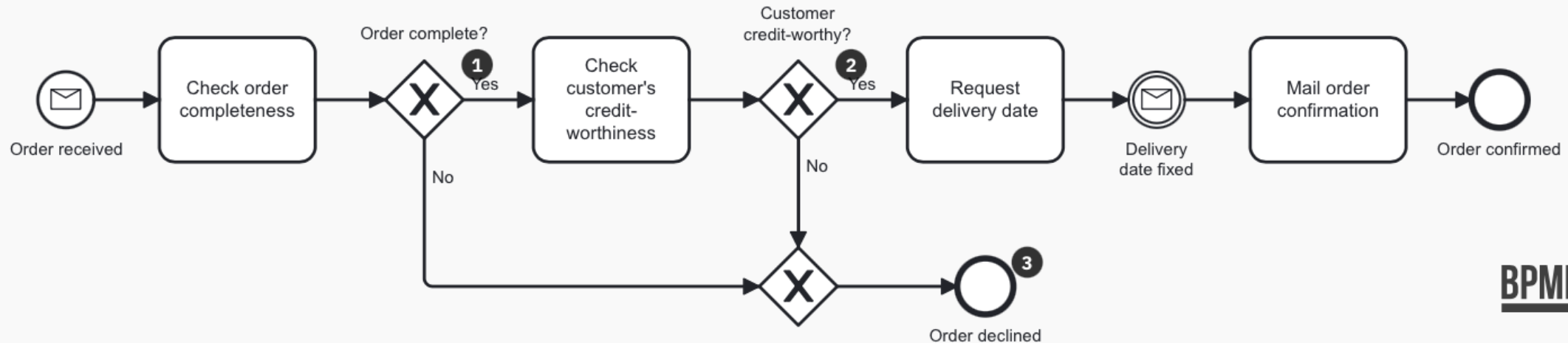
Designing a Business Process

Model beyond
the **happy path**

1) The *exclusive gateway* deals with the potentially problematic result of incomplete order data.

2) Preceding task already dealt with the actual work of checking customer's creditworthiness. Use data-based gateway to move along.

3) The *end event* characterizes the undesired end result "order declined," which we now reach because of having modelled two problems.



Hands On Exercise



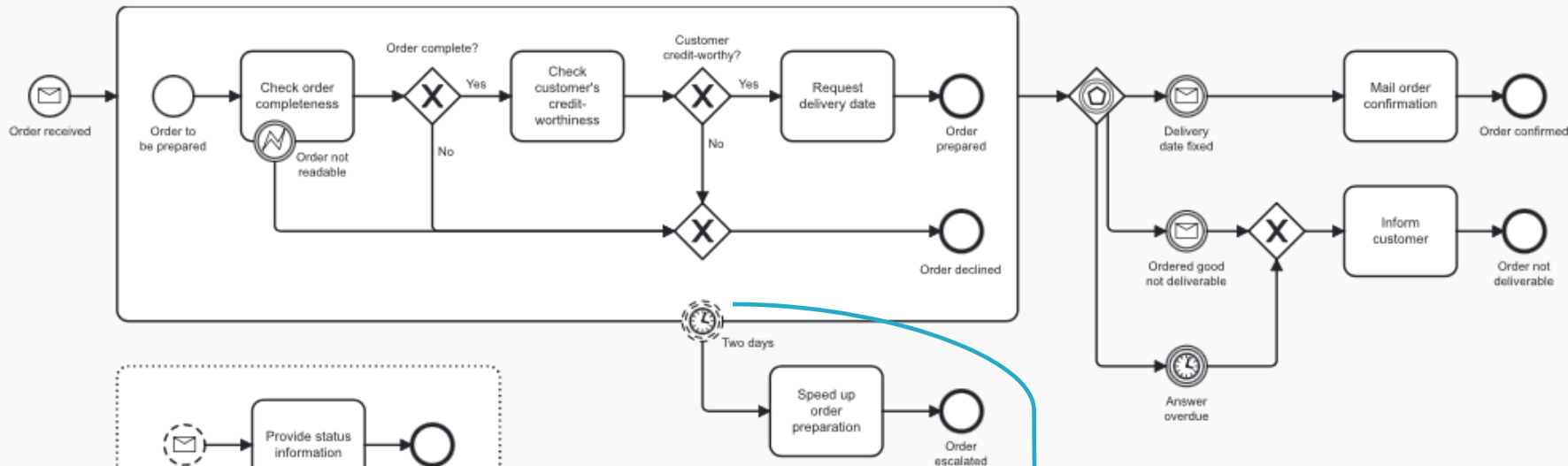
- Complete Camunda Local Environment Setup
- Lead To Opportunity: Business Process

Recap

- Defining and Deploying Business Process
- Executing process instance
- Looking at the execution flow
- Service Task (or Worker)
- Distributed data processing



Designing a Business Process



The *non-interrupting start message event* of the event subprocess

The *interrupting start message event* of the event subprocess that the customer requests cancellation

The *non-interrupting boundary timer event* allows us to speed up order preparation

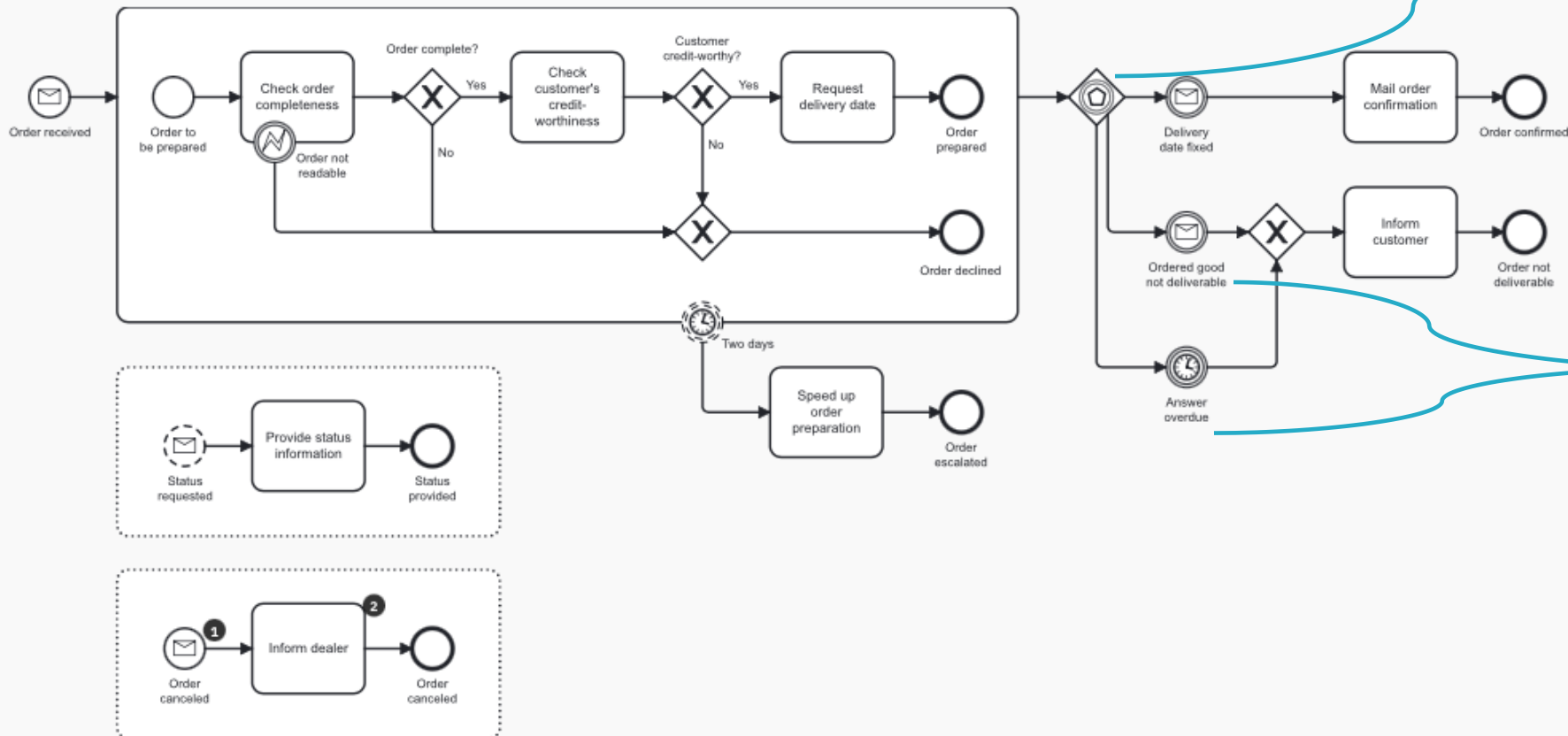
Interrupting
Versus
Non-Interrupting

Use **events** and **timers** to build flexibility

- Boundary events to add activities on triggers



Designing a Business Process



Continuing with previous slide:

Use an event-based gateway to passively wait for what happens next.

The *intermediate message event / timer event* allows us to deal with the undesired situations as outlined here.



Connectors

Protocol Connectors

- REST API
- SOAP
- GraphQL
- HTTP Webhook

Developed by
Camunda or
Partner or
Community

Outbound Connectors

- Amazon SQS
- SQL
- Salesforce
- Teams
- WhatsApp

Inbound Connectors

- RabbitMQ
- Slack
- Amazon SQS

- A Connector is a reusable building block that performs the integration with an external system.
- **Outbound Connectors** allow workflows to trigger external systems or services, making it possible to integrate workflows with other parts of a business process or system architecture.
- **Inbound Connectors** enable workflows to receive data or messages from external systems or services



Decision Model Notation (DMN)

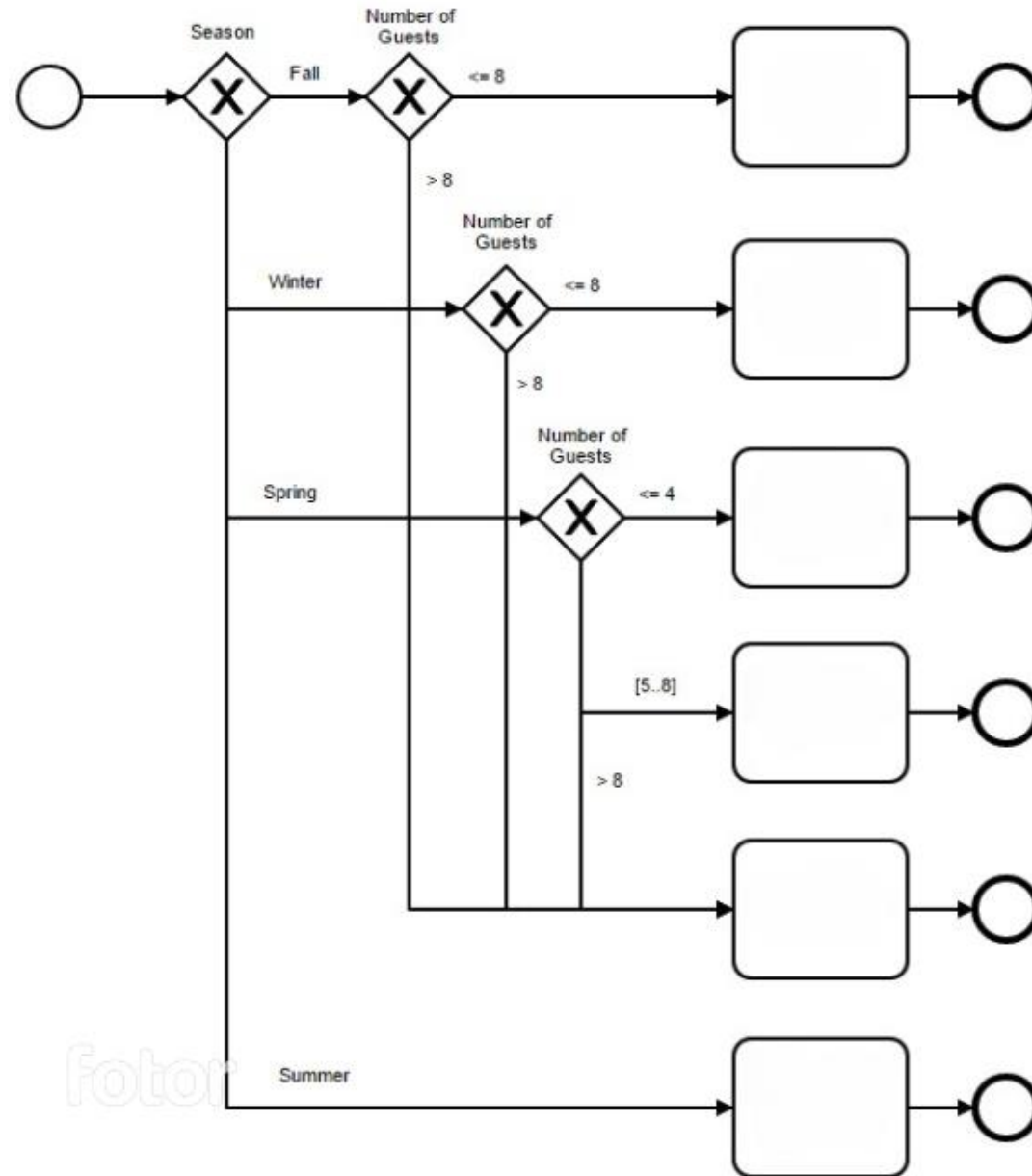


Why DMN?

Let's do an exercise !

Preparing a dish based on

- Season and
- Number of guests



Overview

- DMN is managed by **OMG** – a Standards Development Organization, which has established other world-wide standards, e.g., BPMN and UML
- The DMN standard is supported by several software vendors
- Easy-to-read tables that can be executed directly by a decision engine (like Camunda)
- Allows rapid changes to the rules via simple edits



Decision Table

- Input Data
- Literal Expression
 - Mathematical functions
 - String functions
- Rule representation
- Hit Policy

Hit Policy	XML representation
Unique	UNIQUE
Any	ANY
First	FIRST
Rule order	RULE ORDER
Collect	COLLECT

Dish Decision Hit Policy: Unique				
	When	And	Then	
	Season	How many guests	Dish	Annotations
	"Winter","Summer","Spring"	number	string	
1	"Winter"	<= 8	"Potato Soup"	
2	"Winter"	> 8	"Pasta"	
3	"Summer"	> 10	"Light Salad"	
4	"Summer"	>= 10	"Beans Salad"	
5	"Spring"	< 10	"Stew"	
6	"Spring"	>= 10	"Stew"	
+	-	-		

Specifies what the results of the evaluation of a decision table consist of.



Decision Table

Hit Policy	XML Representation	Output Structure	Description
Unique	UNIQUE	0 or 1	Only a single rule can be satisfied or no rule at all. If more than one rule is satisfied, the Unique hit policy is violated.
Any	ANY	0 or 1	Multiple rules can be satisfied. However, all satisfied rules must generate the same output.
First	FIRST	0 or 1	Multiple rules can be satisfied. The decision table result contains only the output of the first satisfied rule.
Rule Order	RULEORDER	0 or 1 or Many	Multiple rules can be satisfied. The decision table result contains the output of all satisfied rules in the order of the rules in the decision table.
Collect	COLLECT	0 or 1 or Many	Multiple rules can be satisfied. The decision table result contains the output of all satisfied rules in an arbitrary order as a list.



Hands On Exercise

- Add a Decision Table to the process definition: Lead To Opportunity



Decision Table

- Aggregator supported for Hit Policy COLLECT
- SUM
- COUNT
- MIN
- MAX

For example, the employee has been working in the company for 3.5 years. So the first, second and third rule will match and the result of the decision table is 600, since the output is summed up.

SUM aggregator

The SUM aggregator sums up all outputs from the satisfied rules.

Salary		Hit Policy: Collect (Sum) ▾
	When	Then
	Years at company integer	Bonus integer
1	> 1	100
2	> 2	200
3	> 3	300
4	> 5	500
+	-	



End of Day 2

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